



**CAGAYAN STATE UNIVERSITY
CARIG CAMPUS**

**COLLEGE OF ENGINEERING
AND ARCHITECTURE**

1

**DEVELOPMENT OF AN AUTOMATED WORKFLOW DATABASE
SYSTEM FOR CONSTRUCTION PROGRESS MONITORING AND
EMAIL NOTIFICATION IN PROVINCIAL GOVERNMENT OF
CAGAYAN**

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BACHELOR OF SCIENCE IN CIVIL ENGINEERING

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PROPOSAL SUMMARY

This study focuses on the development of an automated workflow database system for construction progress monitoring and email notification in the Provincial Engineer's Office (PEO), Construction Division of the Provincial Government of Cagayan. The study aims to address concerns associated with conventional construction monitoring and reporting practices, including manual data processing, paper-based records, individual spreadsheet files, and information stored in different locations. The proposed system, ConstructFlow, will centralize project information and automate selected monitoring, reporting, scheduling, and notification processes. The study will employ a Design and Development Research (DDR) approach using mixed methods. Data will be gathered through structured questionnaires and semi-structured interviews from eleven (11) purposively selected respondents representing Engineer I, Engineer II, Engineer III, Engineer IV, and Contractor roles. The system will incorporate weekly progress reporting, Precedence Diagramming Method (PDM) scheduling, S-curve analysis, Bar chart visualization, PDF progress reports, and automated email notifications. ConstructFlow will be evaluated in terms of usability, reliability, functionality, and performance efficiency, while controlled test cases will be used to compare automated and manual reporting in terms of accuracy and efficiency.



TABLE OF CONTENTS

APPROVAL SHEET	2
LIST OF FIGURES	6
CHAPTER I	1
INTRODUCTION	1
1.1 Background of the Study	1
1.2 Statement of the Problem	4
1.4 Hypotheses of the Study	8
1.5 Conceptual Framework of the Study	9
1.6 Significance of the Study	11
1.7 Scope and Delimitation	13
1.8 Definition of Terms	14
CHAPTER II	16
REVIEW OF RELATED LITERATURE AND STUDIES	16
2.1. Introduction	16
2.2. Main Topic	17
2.3. International Perspective	32
2.4. Local Perspective	38
2.5. Synthesis	41
CHAPTER III	43
METHODOLOGY	43
3.1 Study Area	43
3.2 Research Design	45
3.3 Materials, Instruments, and Respondents	48
3.4 Population and Sample	50
3.5 Data Gathering Procedure	52
3.6 Reliability Test	54
3.7 Statistical Tools	56
3.8 Ethical Consideration	60
REFERENCES	62
APPENDICES	73
Appendix A. Pre-test Questionnaire	73
Appendix B. Post-test Questionnaire	77



LIST OF TABLES

Table No.	Title of Table	Page
1	Components of the Survey Questionnaire	48
2	Distribution of respondents by construction project	49
5	Cronbach's Alpha Interpretation	54
6	Cohen's d Interpretation	57



LIST OF FIGURES

Figure No.	Title of Figure	Page
1	Conceptual Framework of the Study	9
2	Map of Cagayan Provincial Capitol	43
3	Research Design Framework	46
3	Project Dashboard Overview	52



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CARIG CAMPUS**

**COLLEGE OF ENGINEERING
AND ARCHITECTURE**

7

ABSTRACT

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GOVERNMENT OF CAGAYAN

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BACHELOR OF SCIENCE IN CIVIL ENGINEERING



CHAPTER I

INTRODUCTION

1.1 Background of the Study

The construction industry is an essential part of the development of infrastructure and public utilities; however, poor progress tracking and reporting are still making it hard for projects to succeed everywhere. Unless research on these problems is done, construction projects continue to face the risk of delays and slow progress reporting procedures. Internationally, delayed reporting and fragmented workflows pose challenges to stakeholders to detect problems timely, leading to the increased project schedules and decreased productivity. Even though automated systems have been presented, their minimal practical adoption and application limits the effectiveness of the systems in real-life contexts (Qureshi et al., 2023). Moreover, the use of new and expensive technologies like laser scanners and camera-based systems is expensive and demands a high level of technical knowledge which may not be easily accessible by most organizations. Also, the cost of the advanced technologies are high, and their usage needs a lot of technical expertise; thus, many organizations cannot easily utilize them. The construction industry will not be able to overcome these weaknesses unless studies are conducted to overcome them, impacting the performance of projects and their satisfaction to stakeholders.



Construction projects in the Philippines are on the rise because of the growth in the number of infrastructure requirements. Nevertheless, a large number of organizations continue to use manual means of progress monitoring. Such practices usually result in delayed reporting which greatly influences communication between stakeholders of the project. Consequently, such critical aspects as design modifications, unfavorable weather conditions and slow decision making are not tackled promptly leading to project delays and higher costs (Amoah et al., 2024). The scope creep intensifies these problems because it happens when unexpected changes make the project requirements to be more than what was initially planned. This makes it complicated to track progress. Although automated monitoring tools exist, their usage is low because of the high deployment cost and complexity of the system. This underscores the importance of a more practical, accurate, cheaper, and accessible to use solution that is localized.

The Cagayan Provincial-Capitol Provincial Engineer's Office (PEO), Construction Division is an important role player in the governance and infrastructure development of the Province of Cagayan. The PGC-PEO, Construction Division is tasked to manage the implementation, monitoring and evaluation of different Government infrastructure projects which contribute to the socio-economic development of the Province. Activities in the PGC-PEO, Construction Division, are representative of those found in government offices in semi-urban and rural settings where problems of common concern are prevalent. The challenges faced are ongoing manual processes, restricted digital integration, and project monitoring and reporting delays due to disjointed processes and communication. The limitations can impact on the accuracy, timeliness and effectiveness of decision



making, especially in infrastructure projects that require up to date and accurate information for appropriate management and supervision of the project.

This research is being done to fill significant limitations and gaps in research of existing construction progress monitoring systems. Although some previous studies have been conducted on advanced technologies such as Computer Vision (CV) and automated monitoring solutions, most of them merely focus on individual aspects of the work such as data capture or progress visualization and do not integrate essential project management processes into a unified workflow (Gharib & Moselhi, 2023; Qureshi et al., 2023). Moreover, current solutions are not feasible to small enterprises with limited resources as they often require expensive hardware and complicated technical configurations. Apparent is the lack of research carried out on the creation of a web-based automated workflow database system that is cost-effective and that consolidates data input, progress computation, scheduling methods like Precedence Diagramming Method (PDM), S-curve analysis, Bar charts, critical path calculation, and automated email reporting into one platform. The study will address that gap by designing and developing an integrated and user-friendly system. Through it, it improves productivity, communication, accuracy, and decision-making in construction project management by presenting a new strategy, which has not been well discussed in the previous literature: an automated workflow database system with reporting and notification capabilities.



1.2 Statement of the Problem

Construction project progress monitoring and reporting are essential activities to ensure that projects are completed within the planned schedule, budget, and scope. However, many construction organizations still rely on manual methods of monitoring, recording, and reporting project progress, which may lead to delayed information dissemination, project information being stored in different files, and slower decision-making. While various automated construction project monitoring systems have been developed, many of these solutions use advanced technologies such as laser scanners, and Computer Vision (CV)-based systems that require significant investments in hardware, software, and technical expertise (Qureshi et al., 2023; Shamsollahi, 2024). As a result, their implementation may not be practical for many government offices and organizations with limited resources.

Similarly, commercially available project management software such as Primavera P6 and Microsoft Project offer powerful scheduling and reporting capabilities but often require subscription fees, specialized training, and technical knowledge to fully utilize their features. Consequently, many organizations continue to rely on commonly available tools such as Microsoft Excel and Microsoft Word for project documentation, reporting, and data management. Although these tools are accessible and familiar to users, they do not provide integrated features for centralized project monitoring, automated progress calculations, scheduling analysis, and systematic report dissemination.



In the Province of Cagayan, where construction and infrastructure projects continue to increase, project monitoring and reporting activities are still largely carried out through manual workflows. Delays remain one of the major challenges in construction projects and are often caused by changes in project plans, insufficient communication among stakeholders, adverse weather conditions, delayed material deliveries, and slow decision-making processes (Amoah et al., 2024). These challenges may result in cost overruns, schedule delays, reduced productivity, and, in some cases, project termination. Furthermore, delays in the submission and dissemination of progress reports may prevent project managers and stakeholders from identifying issues early and implementing corrective actions on time.

The lack of an integrated workflow management system further contributes to inefficiencies in project monitoring and reporting. Project information is often stored in multiple files and documents, while progress calculations may be performed using spreadsheet formulas and manual data encoding. However, these processes are typically carried out across separate files and systems, requiring considerable time and effort in data consolidation, verification, and report preparation. In addition, communication among project personnel frequently depends on manual follow-ups, which may delay the delivery of important project updates and decisions. These limitations make it difficult to maintain accurate, timely, and centralized project information.

Although existing studies have primarily focused on automated construction progress monitoring and reporting, limited attention has been given to the development of an



affordable and integrated workflow system that combines project data management, scheduling functions, progress calculations, reporting, and automated email notifications within a single platform. This research gap highlights the need for a practical solution that can support the daily monitoring and reporting requirements of construction organizations without requiring expensive technologies or extensive technical expertise.

Therefore, this study seeks to develop a web-based automated workflow database system for the Construction Division of the Provincial Engineer's Office, Provincial Government of Cagayan. The proposed system aims to provide centralized project data management, automated progress computation, Precedence Diagramming Method (PDM) scheduling, S-curve analysis, Bar chart visualization, and systematic email notifications for project reports. By integrating these functions into a single platform, the study aims to improve the efficiency, accuracy, decision-making, coordination and transparency among the stakeholders.



1.3 Objectives of the Study

The study aims to develop a web-based automated workflow database system with email notifications for construction progress monitoring and reporting for stakeholders of the Construction Division of the Provincial Engineer's Office, in accordance with their workflow process.

The objectives of the study are the following:

1. To determine the system and reporting format used by the Provincial Engineer's Office, Construction Division in construction progress monitoring.
2. To design and develop a web-based automated workflow database system to monitor progress and provide stakeholder reporting with email notification, Precedence Diagramming Method (PDM) scheduling, S-curve analysis and Bar chart.
3. To implement and test the system's usability, reliability and functionality.
4. To compare the accuracy and efficiency of the automated reporting system with the existing manual reporting process through controlled test cases.



1.4 Hypotheses of the Study

1.4.1 Null Hypothesis (H_0):

H_{01} : The developed automated workflow system's usability has no significant effect on the level of its users' satisfaction.

H_{02} : The developed automated workflow system's reliability has no significant effect on the level of users' satisfaction.

H_{03} : The developed automated workflow system's functionality has no significant effect on the level of users' satisfaction.



1.5 Conceptual Framework of the Study

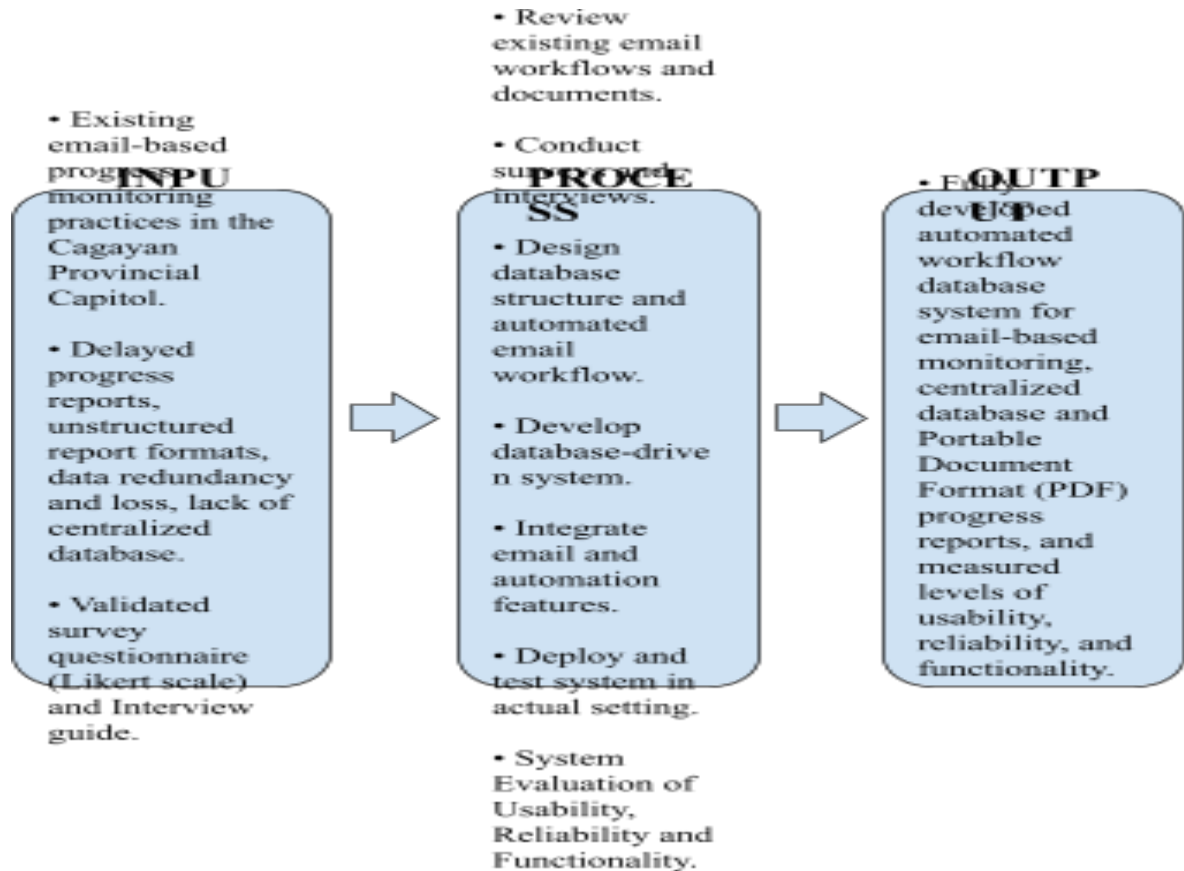


Figure 1. Conceptual Framework of the Study

This study adapts the Input–Process–Output (IPO) model as a framework to develop and analyze an automated workflow database system with email based notifications to monitor progress in the PGC-PEO, Construction Division. The framework shows the transformation of the raw project data and user requirements to the automated monitoring and reporting system through the system development processes. The study assumes that the efficiency of monitoring project progress via email may be enhanced by converting traditional workflows based on emails to an automated database-driven system. System design, development, and evaluation stages process the inputs of the project



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10

reports, schedules, stakeholder requirements and existing workflow issues. The result of these processes is the development of an integrated system that automates reporting and workflow among the stakeholders. The output of this is a functional system that will enable real-time monitoring, standardized reporting, and enhanced decision-making support to project management operations.



1.6 Significance of the Study

This study aims to develop an automated web-based workflow for construction progress monitoring and stakeholder reporting at the Provincial Government of Cagayan-Provincial Engineer's Office, Construction Division. This research is deemed significant to the following entities.

PEO Engineer IV. The study will enhance the checking and monitoring of weekly approved progress reports submitted via the system. The automated workflow database system will help facilitate accurate reporting, less manual consolidation of documents, and faster report submission to the administrator of verified reports.

PEO Engineer III. This study offers an avenue to monitor project progress, review reports, and make decisions in a timely manner. It simplifies the manual work, reduces time wastage in enhancing reports and supervision with real-time updates and automatic email notifications.

PEO Engineer II. The automated workflow database system will enable accurate and organized reporting, reduce manual checking and assist in the fast validation and processing of project progress reports.

PEO Engineer I. This study provides a more effective method of submitting weekly progress reports. The automated workflow eases documentation process, minimizes repetitive work of administration and has ensured that the submitted information is checked by the PEO Engineer II and to be reviewed and approved by PEO Engineer III.



Contractor. The study will help provide an organized system sequence of sending out the project updates like S-Curve, Bar Chart and Precedence Diagramming Method (PDM). The automated workflow database system will keep contractors on track with project progress, limit manual reporting, decrease the chance for errors and aid in coordination with the Provincial Engineer's Office.

Provincial Engineer's Office. This study leads to more effective project monitoring and supervision. By enabling project data to be accessed accurately and in time, the system enhances efficiency, accuracy, and decision making which is critical in the management of the projects. It improves the organization of data and record maintenance, which results in improved productivity and low chances of lost documents.

The Researchers. This study supports their future study on creating an automated workflow with email notifications to track construction progress and stakeholder reporting in Provincial Engineer's Office.



1.7 Scope and Delimitation

This study focuses on an automated workflow database system using email notifications to monitor construction progress and report to stakeholders in the Construction Division of the Provincial Engineer's Office (PEO), Provincial Government of Cagayan. It specifically discusses the design, development and implementation of a web-based system with an integrated database that automates data processing and provides Precedence Diagramming Method (PDM) scheduling, S-curve analysis, Bar charts and email notifications to targeted project stakeholders. Area Engineers are still required to collect progress data that will be inserted into the system and used to calculate progress, generate reports and email notification. Also, the study does not use more advanced technology like laser scanners, camera-based and computer vision-based progress monitoring. The scope is limited to Contractors, PEO Engineer I, PEO Engineer II, PEO Engineer III, and PEO Engineer IV involved in specific construction projects. The office staff manually forwards the generated PDF progress reports of the system to the Provincial Engineer for reviewing and decision-making. Progress reporting is done on a weekly basis. Additionally, the system will require internet connection to fully operate. This study will assess the effectiveness of the system to enhance efficiency, accuracy decision-making, coordination and transparency among the stakeholders.



1.8 Definition of Terms

Automated Workflow System - The web-based collaborative environment is used to automate data entry of Cagayan Provincial Capitol construction progress based on the daily activity at the site, automatically adjusting the Precedence Diagramming Method (PDM) schedules, estimating the S-curve progress percentage, Bar chart, and creating PDF-based reports that are emailed to stakeholders in real time.

Bar Chart - A bar chart representation in the Cagayan Provincial Capitol, Provincial Engineer's Office, Construction Division, automated workflow system visualizes task schedules, task durations, task dependencies and progress updates based on proven daily site data, automatically integrating Precedence Diagramming Method (PDM) schedules, S-curve analysis to track delays (e.g., typhoons, material shortages, permits) and generate real-time reports for Cagayan Provincial Capitol projects.

Computer Vision (CV) - Artificial Intelligence (AI) that is able to independently track the project progress on-site through analysis of images is not regarded in this study since it takes up a significant portion of hardware.

Construction Project Monitoring - An web-based system of tracking the daily accomplishments, delays and productivity of the water system projects of Cagayan Provincial Capitol in Cagayan, Philippines, to address typhoon-impacted projects, shortages in materials and permit delays.



Database - A central database where Cagayan Provincial Capitol construction project information such as weekly progress reports which consist of PDM schedules, S-curve values, Bar chart timelines. It provides one source of transparency to all users, allows a simple search and comparison among various Cagayan Provincial Capitol projects, offers real-time visibility of project status, and also allows full audit trail of all project status updates and decisions.

Precedence Diagramming Method (PDM) - The Precedence Diagramming Method is used to prepare the construction task dependencies, using four relationship types; Finish-to-Start (FS), Start-to-Start (SS), Finish-to-Finish (FF) and Start-to-Finish (SF). This diagram is automatically updated when there are delays in water district projects of the Cagayan Provincial Capitol.

Progress Documentation - Site photos, measurements, daily accomplishments of Cagayan Provincial Capitol construction projects are stored digitally, and this data can be used to track S-curve progress, Bar chart and calculate percentage completion.

S-Curve Analysis - Graphical tool that represents the cumulative planned and actual progress of the project life to estimate the date of completion and analyze the variances and performance of a project in Cagayan Provincial Capitol projects.

Slippage - Scheduling impacts (flooding, typhoons, delivery route delays, permit applications) are captured, categorized, and updated automatically to be immediately included in the PDM schedule, S-curve analysis and Bar chart in the Cagayan Provincial Capitol workflow system.



CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1. Introduction

The rapid advancement of information and communication technologies has significantly transformed the operations of organizations, especially in tracking and reporting workflows. In this research study, the suggested system, Development of an Automated Workflow Database System for Email-based Progress Monitoring in PGC-PEO, Construction Division, refers to the design and implementation of a digital platform that integrates task tracking, database management, and automated email notifications to streamline progress monitoring processes. The main purpose of this paper is to explore literature reviews and other related studies that can help establish the system, especially in the areas of workflow automation, database systems, and email-based reporting mechanisms. This section aims to provide readers with a comprehensive understanding of the concepts, frameworks, and technologies that underpin the study, as well as to identify relevant findings that inform the system's usability, reliability and functionality.



2.2. Main Topic

2.2.1 Construction Progress Monitoring

Construction progress monitoring is an essential factor in the construction industry (Gharib & Moselhi, 2023). Through this, engineers and managers are able to monitor project expenses and schedules, making it possible to determine whether the project is running according to the budget and the stipulated period completion (Ekanayake et al., 2021). With this process, management has the ability to act immediately to rectify any deviations. Moreover, construction monitoring is a process through which data is systematically collected, documented, and evaluated in order to determine the present condition. This process includes site inspections, measurement of completed work, and updating records (Vassena et al., 2023). Comparing the projected time frame with the actual construction progress enables the parties concerned to establish whether the construction is proceeding as per plan (Puri & Turkan, 2020).

In complex construction processes, monitoring becomes essential as it helps identify issues such as unexpected expenses and rework. Additionally, efficient monitoring improves decision-making, coordination, and transparency among stakeholders, ensuring the timely completion of construction projects and reducing the risk of delays and cost overruns (Oni, 2025).



2.2.2 Construction Project Scheduling Techniques

Construction projects have become increasingly complicated due to the expansion in scale and impact of technology on the field. Therefore, more advanced methods of scheduling should be used to ensure that construction projects are completed on time, in the budget, and within the limits of project scope. Several project management methods that can enhance construction projects' performance have been identified, including the use of bar charts, S-curve analysis, and Precedence Diagramming Method (PDM) (Parlak, 2016).

Bar charts, commonly referred to as Gantt charts, are some of the most important methods used in scheduling construction projects. According to Wadhwa (2024), they show tasks and their dependencies, helping managers organize work in order and reduce delays. The main elements of bar charts include activities, duration, dependencies, bars, and milestones. Some bar charts may include information about the resource allocation required to complete project activities.

On the other hand, an S-curve is a graph showing the relationship between the duration of construction projects and the amount of work completed, budget spent, or labor input. The method is extensively applied in public and private sectors to control schedules and assess project performance (Irawan et al., 2023). By comparing the actual S-curve with the planned curve, deviations such as delays or early completions can be identified immediately, allowing timely corrective actions to maintain project objectives.



The term S-curve analysis involves assessing the characteristics of an S-curve in construction projects to examine their performance and effectiveness. The method allows project managers to understand whether the construction activities are implemented in accordance with the initial plan, determine productivity trends, and take appropriate actions to minimize the risk of cost overruns, delays, and productivity loss.

Finally, Precedence Diagramming Method (PDM), otherwise known as Activity on Node (AON), is a scheduling method applied in the field of construction to plan and visualize project activities. According to Novitasari et al. (2018), PDM uses nodes to represent activities, and links between them define the order in which the activities should be completed. Additionally, it uses four types of constraints: Start-to-Finish, Start-To-Start, Finish-To-Start and Finish-To-Finish. With these, PDM can accurately represent the dependencies between tasks without requiring dummy activities, which simplifies the project network and makes planning and monitoring more efficient.



2.2.3 Information Systems and Technologies For Construction Project Monitoring

Progress reporting is the foundation of effective construction management. With this, stakeholders are informed about the current status, issues, and future project activities (Mengiste et al., 2025). Weekly reports are essential in assessing discrepancies between planned and actual progress, identifying delays, and adjusting project plans accordingly. Conventionally, weekly progress is prepared manually, requiring site engineers to collect data from multiple sources such as project plans, contracts, site observations, and multimedia records, then compile, analyze, and summarize the information. This manual process involves extensive printing, especially during revisions, is often time-consuming and may be subjective, which can affect the accuracy and consistency of the reports. Because of these limitations, organizations are increasingly exploring ways to streamline reporting activities through digital technologies and workflow automation.

Workflow automation is intended to facilitate repetitive processes, allowing all parties involved to focus on more complex tasks (Ravichandran, 2025). Automation makes the process more transparent and easier to track, which is crucial in meeting high requirements in terms of reporting and compliance with regulations. In relation to construction project monitoring, it means that reports will be produced promptly and reliably. However, effective automation relies on the availability of organized and accessible project information, highlighting the importance of database systems.

Database systems play a significant role in construction project management as they help manage the large volume of data generated throughout a project (Tanga et al.,



2022). The construction industry generates substantial amounts of data through progress monitoring that need to be shared. With this, database systems become essential for the efficient storage, retrieval, and management of information (Deng, 2024). These systems allow the organization of important project data such as schedules, resource allocation, costs, and progress reports, enabling stakeholders to access accurate and timely information for decision-making.

Furthermore, databases help ensure data consistency, reduce data redundancy, and ensure data integrity. These factors are very crucial in ensuring that project data remains consistent. However, despite all these advantages, most construction firms have yet to embrace the use of such systems and continue using manual methods like spreadsheets, word processing packages, and paperwork (Toh et al., 2016).

To address these challenges, the integration of database systems within automated workflows offers a more efficient approach to construction project data management. The database system offers a central mechanism through which project information can be recorded and retrieved, whereas an automated workflow system minimizes manual activities by automating data processes based on pre-set rules. In general, the use of a central database system and automation in workflow can result in improved accuracy, efficiency, and transparency of data. While efficient data management and automation are essential, timely dissemination of information to stakeholders is equally important in construction project monitoring.



Email continues to be one of the most frequently used means of communication in businesses (Demiroglu, 2020). It is extensively utilized in various sectors because of the speed, efficiency, and documentation of communication regardless of the stakeholders' geographical locations (Smith Bailey, 2018). Email allows files and reports to be shared directly, which makes communication more flexible and organized in project settings. Even though email does not have the collaborative features of specialized project management tools, its ability to keep a record of communication makes it useful for sending formal reports and updates to stakeholders.

Beyond email communication, various digital applications have also been introduced to improve project monitoring and reporting in the construction industry. These include construction project management software and cloud-based platforms that support project scheduling, monitoring, reporting, document management, and stakeholder coordination. Among the commonly used applications are Oracle Primavera P6, Microsoft Project, Procore, and Autodesk Construction Cloud. Market analyses indicate that Oracle, Procore, and Autodesk are among the leading providers of construction management solutions globally (Fortune Business Insights, 2026). Likewise, Primavera P6 and Microsoft Project are frequently discussed in construction project management studies because of their capabilities in project scheduling, monitoring, and control (Gajera, 2024).

One of the most widely used construction project management software applications is Oracle Primavera P6. It is designed to assist project managers in planning, scheduling, monitoring, and controlling construction projects. The software allows users



to create detailed project schedules, allocate resources, establish project baselines, monitor project progress, and generate project performance outputs. Studies have shown that Primavera P6 improves planning and scheduling efficiency, supports resource management, and enhances project monitoring through progress tracking and schedule control (Kumar et al., 2017). In addition, the software supports simultaneous management of work breakdown structures for multiple large-scale projects (Kohli, 2017).

Despite these advantages, Primavera P6 also has several limitations. According to Parlak (2016), the software requires specialized training and technical expertise because of its complexity, which may make it difficult for inexperienced users to utilize effectively. The cost of software licensing may also be higher than many competing project management applications. Furthermore, some of its features may not be necessary for all organizations, while its many functions may make it harder to learn and require more effort to manage. Although Primavera P6 provides comprehensive project scheduling and control capabilities, its complexity and cost may affect its suitability depending on the specific needs and requirements of an organization.

On the other hand, Microsoft Project is a project management software widely used for scheduling, planning, risk analysis, resource management, and report preparation (Nower et al., 2025). It allows project managers to create detailed budgets by calculating costs based on resource rates and task assignments. The software also includes a shared resource pool, which enables resources to be used across multiple projects, with calendars that define their availability. In addition, it supports assigning resources to different tasks



depending on workload and availability, helping improve project coordination and planning efficiency. Microsoft Project is also effective in generating project schedules and presenting plans through Gantt charts. However, Microsoft Project has some limitations. It cannot directly calculate manufacturing output based on raw materials; therefore, additional software may be needed for production-related processes. Nevertheless, newer versions have improved its functions by adding features such as task linking, progress tracking, accessibility improvements, and integration with cloud services like OneDrive and Microsoft accounts (Nower et al., 2025). Furthermore, it shares some limitations with Primavera P6, such as a steep learning curve and high cost. However, Microsoft Project is more file-based and primarily desktop-oriented, which can make sharing and collaboration less convenient compared to more centralized or enterprise-based systems.

Another software is Procore, which offers several advantages in improving project coordination and communication. One of its main features is real-time communication, which allows team members to send messages, comments, and notifications instantly, helping reduce delays in project updates and decisions. It also provides a centralized document management system where project files can be uploaded, accessed, and updated by all stakeholders, ensuring that everyone uses the most recent information and reducing errors caused by outdated documents. In addition, Procore supports task management by allowing project managers to assign tasks, set deadlines, and monitor progress, which helps improve accountability and project organization. The software also includes mobile access, allowing users to view and manage project information on-site and respond to issues in real time. Furthermore, Procore collects project data such as schedules, budgets,



task status, and communications, which can be used for monitoring and reporting purposes. Overall, these features help improve communication, data sharing, and project control in construction management (Gajera, 2019).

Although Procore is effective for construction coordination and field management, it has several limitations in project management functions. It has limited scheduling capabilities compared to dedicated tools like Primavera P6 and Microsoft Project, especially in handling dependencies, critical path analysis, and schedule optimization. Its portfolio management features are also limited, making it less suitable for managing multiple projects with cross-project planning and workload balancing. In addition, it relies on external tools for advanced scheduling, reporting, and financial control. Furthermore, larger organizations may experience longer implementation time and require additional setup and standardization before full use (Tristancho, 2026).

Lastly, Autodesk Construction Cloud (ACC) is a cloud-based construction management software that supports workflows across all stages of construction, from planning and design to construction and daily operations. It enhances collaboration among construction stakeholders such as clients, architects, engineers, and quantity surveyors. In relation to construction project management, ACC demonstrates how digital systems can improve data handling by centralizing project and financial information, allowing documents, drawings, and payment records to be stored and shared in one secure platform, which reduces data fragmentation and delays in reporting (Tian et al., 2026). It also supports real-time cost tracking, change order management, and analysis of updated



project data, improving overall project control and decision-making. In addition, its integration with 5D BIM shows how cost data can be linked with building models to improve estimation, monitoring, and financial control. Although ACC is primarily focused on advanced cost and BIM-based management, its features highlight the importance of integrated and centralized systems in improving construction workflows. However, such systems are often complex, require high implementation cost, and may exceed the needs of organizations that only require basic progress monitoring and reporting functions.

Despite these software solutions, in practice many government engineering offices continue to rely on commonly available applications such as Microsoft Excel, Google Sheets, Microsoft Word, and Google Drive for construction project monitoring and reporting, as these are easier and less expensive to utilize. Microsoft Excel and Google Sheets are commonly used for recording project data and performing progress calculations through predefined formulas, while Microsoft Word is utilized for report preparation and Google Drive for document storage and sharing. Although these applications are widely available, they often operate as separate tools that require manual data transfer, consolidation, and verification of information. As the number of projects increases, managing data across multiple platforms may become inefficient and may increase the risk of data inconsistencies, delayed reporting, and duplication of work. These limitations highlight the need for integrated information systems that combine data management, workflow automation, communication, and reporting functions within a single platform. Such systems can improve efficiency, transparency, and accessibility of project



information, thereby supporting more effective monitoring of government construction projects.

2.2.4 System Usability

Usability is the quality attribute at which a system provides a convenient and easy interface, ensuring that users can achieve outstanding outputs (Keenan et al., 2022). It is regarded as among the critical qualities in developing systems, since usability is the main consideration for measuring success in the service offered by the system (Elberkawi et al., 2016). According to Xie & Matusiak (2016), usability is composed of several key components: learnability, efficiency, memorability, errors and satisfaction. Learnability refers to how easily users can utilize the system upon first use. Efficiency indicates how fast a user can perform tasks in the system. Memorability pertains to how quickly a user can reestablish proficiency after a long period of not using the system. Errors refer to the number of mistakes users make and how easily they can recover from them. Lastly, satisfaction focuses on how content and comfortable the users are with the overall design of the system. Together, these components provide a comprehensive basis for evaluating the overall success of a system.

In addition to the five components, effectiveness may also be considered as another component of usability. According to Russ & Saleem (2018), a system with a high level of usability is expected to be effective, efficient, engaging, error-tolerant, and easy to learn. In general terms, effectiveness refers to accomplishing tasks properly (Patel, 2021). In the context of a web-based system, it refers to how well the system starts and completes



processes in order to produce the intended output. Hence, effectiveness is an important indicator of how accurately and completely a system achieves its intended function.

2.2.5 System Reliability

System Reliability is one of the most critical factors concerning system quality. This factor indicates the capacity of a system to effectively perform tasks without any errors, while ensuring timely and accurate completion. In contemporary information systems, reliability is tightly connected with stability, availability, responsiveness, and error-free operation. According to William H. DeLone and Ephraim R. McLean (2016), system reliability is an important factor influencing user satisfaction and system success. Their theory argues that for a system to be successful, consistency and reliability should always be guaranteed. Petter et al. (2016) discovered that reliable systems, defined by low maintenance and precise processing, significantly enhance user satisfaction. When users experience stable and dependable system performance, their confidence in the system increases, leading to more positive evaluations.

In Library Management Systems and digital information platforms, reliability is an important factor that affects user experience. Islam (2018) found that reliability and accessibility are key factors of user satisfaction in digital library contexts because users depend on steady and accurate access to resources. Additional research on e-learning and web-based applications has produced further evidence to support the findings mentioned above. Almaiah and Alismaiel (2019) proved that the reliability of the system has a significant impact on user satisfaction, arguing that users are less engaged when there are



too many interruptions due to system malfunctioning. Rahman et al. (2020) also found that reliable systems enhance user trust and encourage continued usage, emphasizing the importance of consistent performance in maintaining satisfaction. Cheng (2021) concluded that system reliability improves perceived usefulness and user satisfaction, particularly in online platforms where uninterrupted access is essential. According to Friederich and Lazarova-Molnar (2024), reliability is essential when ensuring efficient performance and reducing downtime in the operation of such systems. The latest findings of Sharma (2024) demonstrate that the higher the availability, the lower the response time, and the fewer errors the system makes, the higher the level of satisfaction from its usage will be. A reliable system ensures accurate data management, timely report generation, and uninterrupted access, which are essential for efficient library operations. Based on the reviewed literature, it is evident that reliability plays a significant role in shaping user satisfaction. Therefore, this study investigates whether the perceived level of system reliability significantly affects user satisfaction, leading to the formulation of the null hypothesis stating that the perceived level of reliability does not impact the level of user satisfaction.



2.2.6 System Functionality

Functionality in systems refers to the ability of the system to perform tasks and provide the necessary features to users (Salleh et al., 2017). This is concerned with how well the system performs its operations and delivers the desired results according to the users' needs. The system should be performing its duties in an accurate and consistent manner.

For a web-based system, the term functionality involves the proper execution of the input provided by the user, command processing, and delivery of the desired result. The functionality of the web-based system is also associated with the completeness of the features of the system and its capacity to facilitate activities it is meant to undertake. A web-based system is supposed to work effectively and produce the desired outputs.

Reja et al. (2022) categorized automated construction monitoring technologies as drone, laser scanner, computer vision, photogrammetry, sensors, and Internet of Things (IoT) technologies. These technologies can make very accurate progress measurement by automatically gathering field data and comparing what has been accomplished with what is scheduled. But, they are primarily concerned with data collection during the construction process and cannot fully solve the problem of project administration, reporting process and organizational communication. The proposed system is complementary to the progress monitoring because of its ability to convert project information gathered during the process into management tools. The automated reporting, centralized documentation, scheduling up and down, email notifications and work-flow



approvals give support to project monitoring, and project administration and managerial decision making.

To enhance the accuracy of monitoring construction progress, Hasan and Sacks (2023) created an integrated monitoring framework based on Building Information Modeling (BIM) and several construction monitoring technologies. They determined that multiple monitoring technologies are more reliable and comprehensive in reporting project status information than a single monitoring technology. But they have a particular emphasis on improving the accuracy of monitoring with BIM and other digital technologies. The suggested Workflow database system is not about several monitoring technologies, but about integration of project management and admin tasks. The system integrates scheduling, database management, progress monitoring, accomplishment reporting, workflow approval and automated email notifications in one system. This ensures that project data is easily accessible to everyone involved, including field engineers, office staff, contractors, and project managers, fostering better coordination, efficiency in project processes, and timely decision-making throughout the project.



2.3. International Perspective

The process of progress monitoring in construction projects is very crucial since it enables an evaluation of the performance of such projects while at the same time ensuring that all activities carried out were done according to plan (Qureshi et al., 2023; Gharib & Moselhi, 2023; Navon, 2019; Omar & Nehdi, 2016). Traditionally, progress monitoring was done manually using inspections and periodic reports. Unfortunately, these processes were inefficient because they were associated with late data provision and sometimes incompleteness (Navon, 2019). Nonetheless, modern times have seen the emergence of new strategies in regards to the application of automation and images for progress monitoring that would guarantee the efficacy and precision of this procedure (Qureshi et al., 2023). Additionally, there has been the inclusion of the concept of Computer Vision (CV) in progress monitoring processes to automate the process of data collection and analysis to ensure real-time progress monitoring (Gharib & Moselhi, 2023). Nonetheless, even though the above-mentioned changes are imperative, they are not widespread because of their complexity and costliness. Consequently, the effectiveness of existing progress monitoring techniques has been improved.

There are various types of scheduling methodologies used in construction such as PDM, Bar Charting, CPM, and S-curve analyses, which are very effective tools for the management of time and resources within projects (Amoah et al., 2024; Kerzner, 2022; Mubarak, 2019; PMI, 2021). These scheduling methodologies ensure that all the critical activities are known, and resources are utilized efficiently, as well as forecasts regarding



completion become more accurate (Kerzner, 2022). Nevertheless, poor coordination along with no synchronization between project progress and scheduling results in project delays (Amoah et al., 2024). (Amoah et al., 2024). Furthermore, scheduling through manual methods leads to out-of-date schedules, which are less useful for making decisions (Mubarak, 2019). Along with that, scheduling manually results in traditional schedules that are less helpful for decision-making (Mubarak, 2019). As can be shown, scheduling can be highly effective when automated techniques and systems are used. The purpose of this project is to include scheduling into an automated workflow system.

It cannot be underestimated how important the role of database systems can be in managing construction project data because such a process requires proper scheduling, progress tracking, and resource allocation (Elazhary et al., 2020; Alaloul et al., 2020; Marzouk & Enaba, 2019). The use of databases ensures easy access, consistency, and reliability of data and allows you for making well-informed progress reports (Elazhary et al., 2020). Despite this advantage, there is a tendency for construction companies to store their data using scattered methods, which leads to inefficiencies and inconsistencies in data (Marzouk & Enaba, 2019). Moreover, automation and real-time reporting capabilities have been limited in the lack of integrated databases (Alaloul et al., 2020). This means that inefficient data management systems hinder its full implementation and delay decision-making processes. This research will therefore emphasize on the importance of adopting an integrated database system within the context of construction management by developing an integrated database-driven system for managing workflows.



According to Shamsollahi (2024), Qureshi et al. (2023), and Alaloul et al. (2020), workflow automation has emerged as an indispensable tool for enhancing coordination and minimizing manual interventions in the process of construction progress monitoring. Automated solutions can reduce human delays by improving data entry, progress tracking, and reporting According to Qureshi et al. (2023). Despite this, existing automation systems have tended to focus on isolated processes instead of the entire workflow process; hence the disorganization of such systems (Shamsollahi, 2024). Furthermore, adoption has been hindered by high costs of installation and complexity (Alaloul et al., 2020). Thus, existing automation technologies are not enough to ensure total workflow management in construction. To solve the problem of inadequate integration of workflow automation systems, this study proposes an automated workflow system comprising several tools in one package.

In order to provide timely reporting and coordination among stakeholders, communication plays a critical role in the success of construction project processes (Alshawhi & Ingirige, 2020; Madanayake, U., Toli, V. (2024). In real-time, update dissemination is possible with the use of email notification systems. Currently, manual reporting approaches are being employed by many organizations, which create misunderstanding, inefficiency, and unsolved issues Madanayake, U., Toli, V. (2024). Also, without having automated notification tools, stakeholders' communication will be less effective (Alshawhi & Ingirige, 2020). Therefore, ineffective communication is associated with poor decision-making and inefficiencies in project management. As an



outcome, this study uses automated email notifications to promote timely and efficient communication as well as automated reporting.

In manual systems, encoding, inspecting, and reporting typically require human intervention, introducing the risk of incomplete records and human errors, particularly when dealing with significant amounts of workflow data. Automated monitoring systems, on the other hand, leverage the power of centralized databases, real-time data processing, predictive analytics, and algorithmic validation to enhance monitoring capabilities and data integrity. Automation has been shown to alleviate workload and increase the consistency of monitoring as well as decrease the possibility of inaccuracies that might occur from manual recording processes (Lotfinejad et al., 2024; Wolfensberger et al., 2024; Tse et al., 2020). Additionally, Wray et al. (2024) reported that automated procedure logging systems resulted in higher accuracy and completeness of logging, compared to manual methods. According to Bandlamudi and Pachala (2025), automated anomaly detection systems can process the large amount of workflow logs effectively, whereas manual methods are not as effective. Furthermore, security and privacy issues such as unauthorized access and data breaches remain a major concern in the use of automated systems, highlighting the critical importance of robust authentication, data integrity, and access control measures (Zhang et al., 2022). Timeliness, data accuracy, and completeness of data are among the critical dimensions of data quality that are hard to achieve in manual systems but can be much easier to achieve using automated systems, as indicated in the literature (Batini & Scannapieco, 2016). The studies as a whole indicate that the automated database workflow systems provide significant advantages over manual



operations, but have limitations in terms of integration, ease of use and scalability, which underscores the need for a more efficient and reliable system for monitoring and reporting on projects.

Technologies for project monitoring in government agencies have advanced significantly, with the adoption of tools such as Computer Vision, Building Information Modeling (BIM), and Internet of Things (IoT) systems to enhance transparency and efficiency (Gharib & Moselhi, 2023; World Bank, 2020; Asian Development Bank, 2021; Shamsollahi, 2024). These technological advances provide better precision in data collection and real-time monitoring of the performance of projects (Gharib & Moselhi, 2023). Nevertheless, many public sector organizations find it hard to implement such methods due to the large amount of financial resources and technical knowledge needed to develop and utilize them (Shamsollahi, 2024). International agencies also highlight the need for affordable, adaptable solutions that are suitable for local government settings (World Bank, 2020; Asian Development Bank, 2021). Therefore, despite being very efficient, the use of such innovations is restricted due to their high cost and complicated nature. This study provides another approach for developing an affordable and user-friendly technology that meets the demand for appropriate monitoring technology in government agencies like the Cagayan Provincial Capitol, Provincial Engineer's Office, Construction Division.

Overall, the studied literature shows that while construction progress monitoring, scheduling strategies, database systems, workflow automation, and communication



technologies have advanced significantly, these elements are frequently created and implemented separately. The creation of an integrated, affordable, and user-friendly automated workflow database system that integrates these features into a single platform still has a significant research need. This study fills this gap by creating a web-based automated workflow system with email reporting that combines data management, scheduling, progress computation, and real-time communication. This offers a complete solution that improves productivity, decreases delays, facilitates better decision-making and communication in building project management.



2.4. Local Perspective

Several local studies focused on one phase of the workflow have shown that Information and Communication Technology (ICT) tools can be useful in monitoring construction projects in the Philippines. An example is Herrera et al. (2025), analyzed the utility of geotagging in the context of infrastructure works in the Philippine Rural Development Project (PRDP). The researchers found out that geotagged photos improved monitoring, accountability, and progress monitoring of the projects by all parties involved in the process, including local governments. Geotagging can be considered to be a good tool for monitoring of infrastructure works, which does not require any proprietary systems or special software tools.. However, the study focused on geographically distributed rural infrastructure projects rather than scheduling-related tools such as PDM and S-curve analysis or stakeholder notification. In order to fill the gaps, this study develops a web-based automated workflow database that integrates with field data for scheduling, updating progress, and reporting for Cagayan Provincial Capitol, Provincial Engineer's Office, Construction Division projects.

ConCheck, an innovative application for the monitoring of construction projects for private organizations, was developed by Pinlac et al. (2023) specifically for use in Pampanga province. This application seeks to overcome many problems, such as communication gaps, delays in the delivery of construction materials and equipment, and accidents that take place during the process. With the help of two surveys and pilot tests on five private organizations engaged in construction work, ConCheck proved beneficial



in improving daily progress through the use of digital reports such as Daily Weather Report, Delay Type, Daily Progress Report, and Accomplishment Report. The study showed that local construction firms are receptive to digital monitoring tools when these tools simplify documentation and improve communication. Nevertheless, ConCheck focused mainly on daily reporting and contractor-level monitoring and did not provide full integration of PDM scheduling, S-curve analysis, or complete workflow automation. Building upon the previous studies' localized approach, this paper proposes a solution framework that is customized to address the Cagayan Provincial Capitol, Provincial Engineer's Office, Construction Division's needs for monitoring and reporting purposes.

In particular, Angulo et al. (2025) created a manpower monitoring system for single-story residential buildings in the region of Pampanga, whose efficiency and usefulness led to high satisfaction among users because of its capability in managing manpower attendance and work progress. Results show that using digital monitoring systems can help enhance staff distribution and minimize delays associated with regional construction projects. While the previous study was concerned with a manpower tracking system, it was not capable of offering automatic scheduling, visual estimation, and e-mail based reporting. Contrary to this, the present research will combine field data encoding, PDM scheduling, S-curve, and emails under one framework.

According to Santos and Jocson (2024), data quality, technical, and skill challenges affect the use of Artificial Intelligence (AI) in the Philippine construction industry. While recognizing the benefits of using Artificial Intelligence (AI) technology to



improve efficiency and reduce the time of construction work, they admit the inability of contractors to implement them owing to the high cost and complexity involved. Therefore, a cheaper and more feasible solution should be found that can be implemented locally. This study proposes a web-based automated workflow system that is easier and more affordable for Cagayan Provincial Capitol, Provincial Engineer's Office, Construction Division to implement than costly Artificial Intelligence (AI) or computer vision-based solutions. Overall, the local literature shows that Philippine construction projects can benefit from digital monitoring tools, geotagging, and web-based reporting applications. However, the studies also reveal a common limitation, most solutions address only specific monitoring tasks and do not provide a complete workflow that combines data entry, scheduling, progress analysis, and automated reporting. This gap justifies the development of the proposed Cagayan Provincial Capitol, Provincial Engineer's Office, Construction Division system, which aims to provide an integrated and practical solution for construction progress monitoring in a local government setting.



2.5. Synthesis

From the literature reviewed, there appears to be a trend toward the adoption of computerization and automation methods in the monitoring of construction project progress. Research conducted internationally focuses on improvements in project performance attained using precise progress monitoring, scheduling, central databases, automation of workflows, and timely communication. Technology such as computer vision, building information modeling, and artificial intelligence, while potentially useful, can be expensive and difficult for smaller or resource-constrained organizations to implement and use effectively.

The same results have also been found in other related local researches involving geotagging, construction monitoring applications, manpower tracking system, and evaluation using artificial intelligence. Generally speaking, technological advancement has improved the supervision, communication, and reporting processes within construction projects. But still, there is a lack of literature and systems that can be used to cover the whole process of construction monitoring from start to finish. The limitations include automation in planning, scheduling, calculation of progress, notification, and storing of data for utility and linear infrastructures.

Based on this literature review, there is still no cost-effective and practical system that combines field data encoding, scheduling, progress monitoring, database management, and automated email reporting in one platform. In line with this, the present study aims to develop a web-based automated workflow database for progress monitoring



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AND ARCHITECTURE**

42

in the Construction Division of the Provincial Engineer's Office, Provincial Government of Cagayan. Through this, the study seeks to contribute an integrated and practical solution for local government construction monitoring.



CHAPTER III

METHODOLOGY

3.1 Study Area

The study will be conducted at the Provincial Government of Cagayan-Provincial Engineer's Office (PGC-PEO), Construction Division. The institution is responsible for the planning, implementation and monitoring of various government infrastructure projects within the province. As one of the main coordinating bodies, it handles wide amounts of project related information, documentation and interdepartmental coordination activities.

The PGC-PEO, Construction Division was chosen as the study area because there is a need for a more integrated and automated system for managing project monitoring and reporting procedures. The existing practices involve manual data processing techniques, such as paper-based records, individual spreadsheet files, and data in various locations. These practices can influence the organization of data, its accessibility, and efficiency in handling project information. The proposed system seeks to resolve these issues by offering a centralized database, automated workflow, and structured reporting tool that will provide a way to make better decision-making and communication.

The need to enhance the organization, accessibility, accuracy, and real-time monitoring capabilities of data in the institution further justify the choice of the study area. The study will aim to offer a practical solution, which will not only improve operational



efficiency but also support decision-making and will act as a role model to other similar government agencies that are faced with similar challenges.

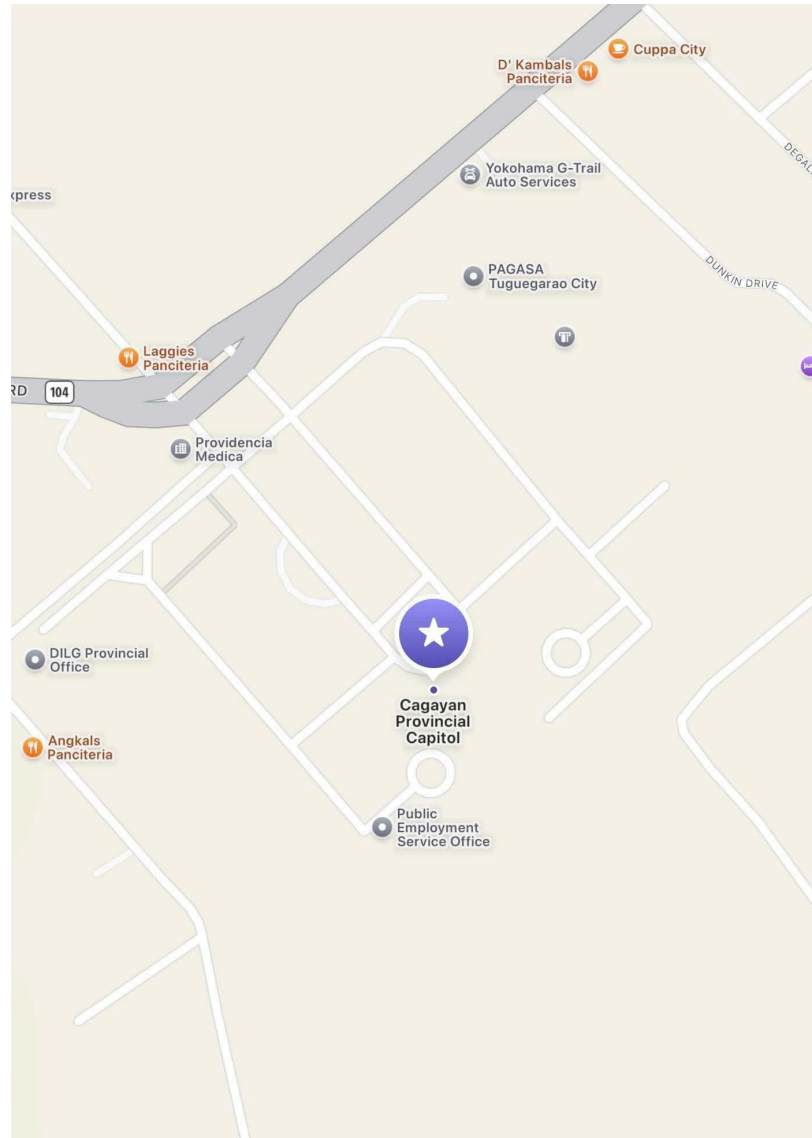


Figure 2. Map of Cagayan Provincial Capitol (Source: Apple Maps)



3.2 Research Design

This study will use a Design and Development Research (DDR) methodology in order to systematically guide the design, development, implementation, and evaluation of a web-based automated workflow system with email reporting for progress monitoring in the Cagayan Provincial Capitol, Provincial Engineer's Office, Construction Division. This research will be a developmental research study in that the researchers will produce a functional system, test it, and validate it through the collection of measurable data. It ensures that the developed workflow system is both practical and effective for real-world application. Design and Development Research (DDR) is appropriate for the study's methodology as it will provide researchers with a systematic way to: create a functional workflow system; implement the workflow system into practice; and evaluate the workflow system's success based on its ability to eliminate workflow problems using the following features: automated e-mail reporting, S-curve analysis, bar graph, and Precedence Diagramming Method scheduling. DDR provides a systematic way for researchers to establish an automated workflow system in a manner that is able to be quantified and verified as being useful in practice.

This study will use a mixed-methods research design, which will allow measurable evaluations of the automated workflow system through quantitative data collected using Likert scale questionnaires and qualitative data collected from user interviews. Using a quantitative approach, Likert scale questionnaires will be administered to assess user perceptions of usability, reliability, functionality, and performance efficiency of the



automated workflow system, providing measurable data for evaluating the automated workflow system.

The qualitative component of the study will use interviews to gain feedback from users about their experiences using the automated workflow system, any difficulties encountered with it, and comparisons of how it compares to manual reporting. The combination of both approaches will provide a more complete evaluation of improvements in workflow by providing both quantitative evidence of system performance and qualitative findings that can be used to explain and support the quantitative data collected. This combination provides a more complete and valid evaluation of the automated workflow system.

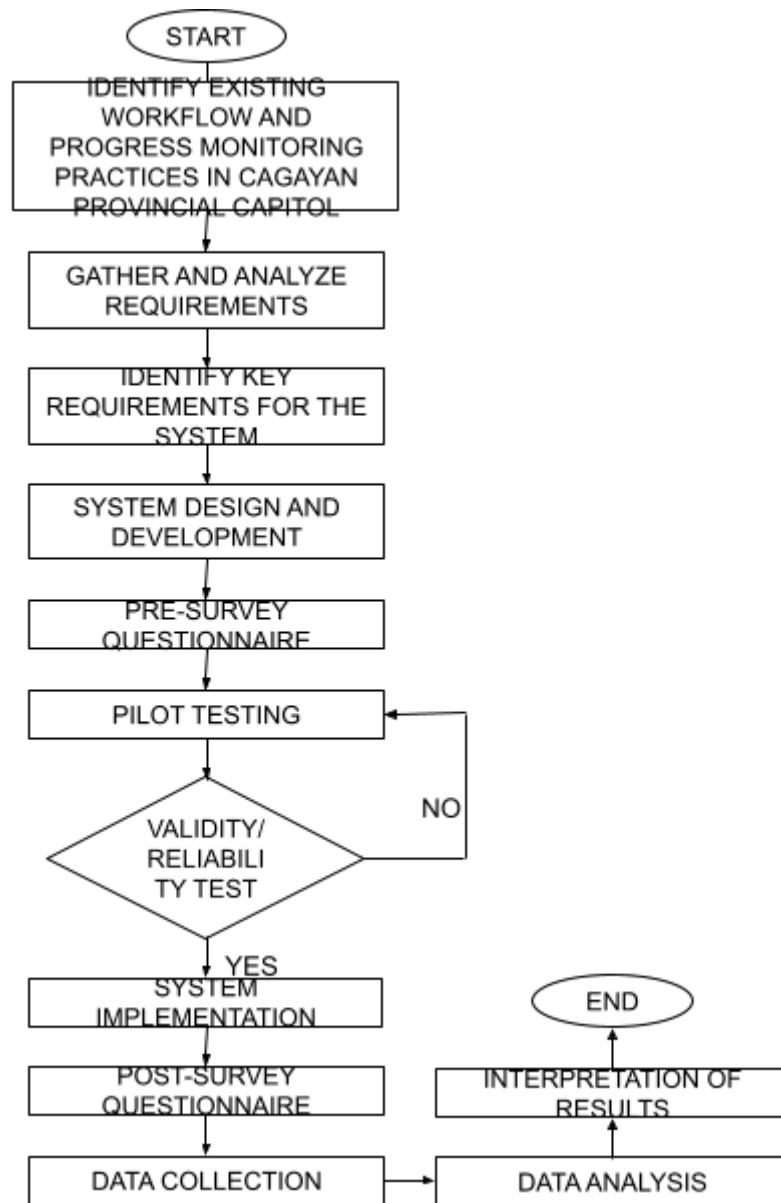


Figure 3. Research Design Framework



3.3 Materials, Instruments, and Respondents

This study will use a web-based automated workflow database that sends emails that are used to track the schedule and progress of construction projects under the Construction Division of the Provincial Engineer's Office of the Provincial Government of Cagayan. The system is designed to perform the Precedence Diagramming Method (PDM) scheduling, S-curve analysis, and Bar chart. It also produces weekly status reports from the system updates submitted by Engineer I. Additionally, the system is accessible via a web browser and requires an internet connection for full functionality.

The database will be used to store construction project data such as the Precedence Diagramming Method (PDM) network of the activity schedules, input of weekly progress reports, percentage of completion, S-curve, and Bar chart.

This study uses a purposive sampling technique in selecting respondents. Purposive sampling is used because the study requires participants who possess specific knowledge, experience, and direct involvement in project monitoring, reporting, and workflow processes within the Cagayan Provincial Capitol, Provincial Engineer's Office. This sampling technique is appropriate because it allows the researchers to gather relevant and reliable information from individuals who are capable of evaluating the developed system based on actual operational experience and system usage. The research instrument used in this study is a structured questionnaire and semi-structured interview questions designed to assess the system's usability, reliability, functionality, and performance efficiency. The components of the survey questionnaire are summarized as shown in Table 1.



Table 1. Components of the Survey Questionnaire

Variable	Information Needed	Nature
Demographic	Gender	Nominal
	Age Group	Ordinal
	Working Experience	Ordinal
User satisfaction	The level of satisfaction with the ability of the system to manage progress monitoring	Ordinal
System usability	The perceptions of users on operability of the system	Ordinal
System reliability	The extent which users perceive the system as reliable while performing its intended functions	Ordinal
System functionality	The perceptions of users on the ability of the system to deliver intended work	Ordinal

Source: Elias and Lubua, 2021

The respondents of the study are Engineer I, Engineer II, Engineer III, Engineer IV, and the Contractor of the selected construction projects handled by the Construction Division of the Provincial Engineer's Office, Provincial Government of Cagayan. These individuals are directly involved in the monitoring and reporting of construction projects, making them the best evaluators of the system.



3.4 Population and Sample

The respondents of this study are the Engineer I, Engineer II, Engineer III, Engineer IV, and the Contractor of the three (3) selected construction projects under the Construction Division of the Provincial Engineer's Office, Provincial Government of Cagayan. A total of eleven (11) respondents will be involved in the study. The participants are selected using purposive sampling based on their direct involvement in the monitoring and reporting of construction progress for the selected projects.

Table 2. Distribution of Respondents for Project 1

POSITION	RESPONDENTS	PERCENTAGE
Engineer I	1	20%
Engineer II	1	20%
Engineer III	1	20%
Engineer IV	1	20%
Contractor	1	20%
Total	5	100%

Table 3. Distribution of Respondents for Project 2

POSITION	RESPONDENTS	PERCENTAGE
Engineer I	1	20%
Engineer II	1	20%
Engineer III	1	20%
Engineer IV	1	20%
Contractor	1	20%
Total	5	100%



Table 4. Distribution of Respondents for Project 3

POSITION	RESPONDENTS	PERCENTAGE
Engineer I	1	20%
Engineer II	1	20%
Engineer III	1	20%
Engineer IV	1	20%
Contractor	1	20%
Total	5	100%



3.5 Data Gathering Procedure

First, the researchers will prepare a formal letter detailing their promise to keep any data collection requests confidential. The Cagayan State University - Carig Campus personnel will review and validate this letter. Second, the researchers will seek the permission to conduct the study and distribute the questionnaire to the respondents from the Cagayan Provincial Capitol, Provincial Engineer's Office, Construction Division head. Once the approval is granted, researchers then prepare and validate the questionnaire. It is then distributed to the respondents through Google Forms or printed copies, depending on their preference and availability. Respondents will be given ample time to respond to the questionnaire. Respondents will be surveyed before and after the pilot testing. Researchers will gather responses, including those recorded from Google Forms. In addition to the survey, researchers will conduct interviews to gain insight into the respondents' experience, system perception, personal experience, and feedback. Respondents will determine the timing of the interview within the confines of the provided questions. Clarifications will be provided by the researchers, and elaborations will be done as a result of follow-up questions. The respondents' requests and feedback will be considered in the further development of the system and construction of the database (Figure 4). The data will determine the system's usability, reliability, and overall functionality.

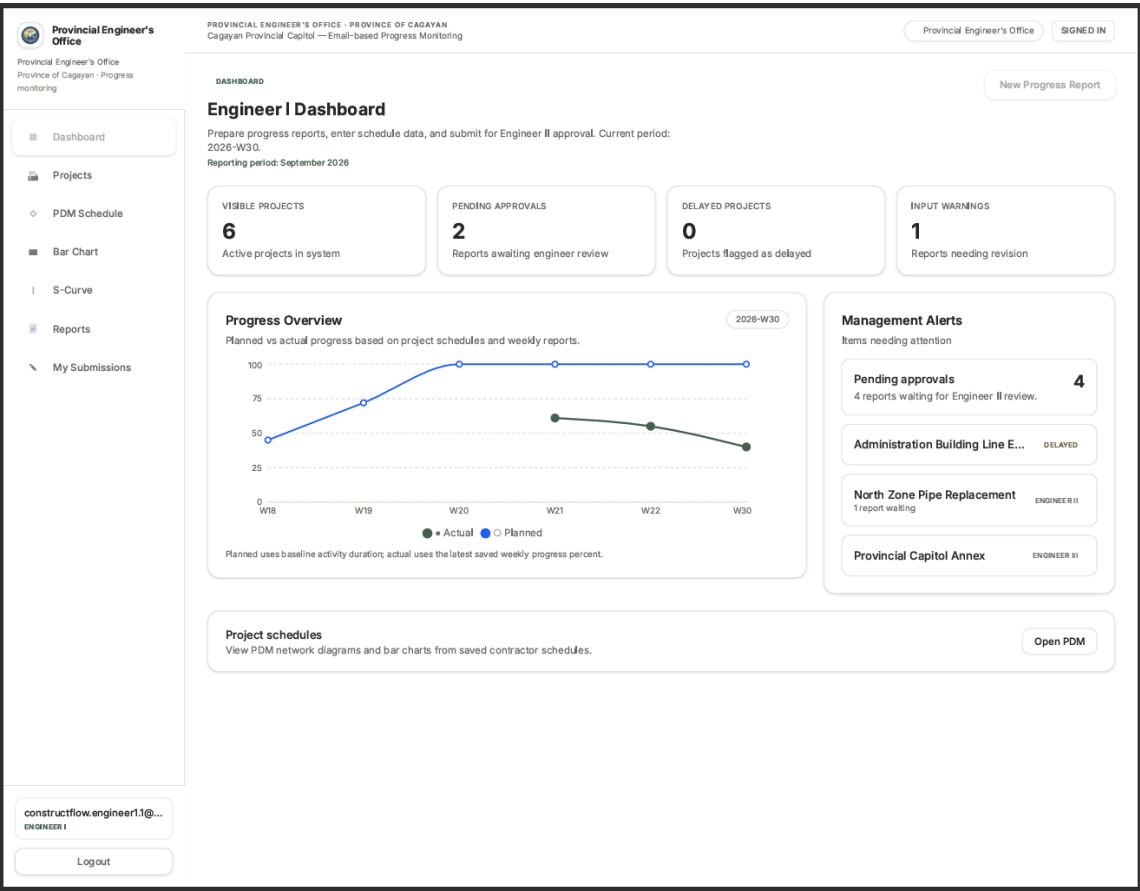


Figure 4. Project Dashboard Overview



3.6 Reliability Test

To measure whether the research instrument is reliable or not, the evaluation of user perception on the usability, reliability, functionality, and performance efficiency of the system will be determined. A structured questionnaire will be used as the primary data-gathering instrument in this study. It will be used for the Engineer I, Engineer II, Engineer III, Engineer IV, and Contractor of the selected construction projects under the Construction Division of the Provincial Engineer's Office, Provincial Government of Cagayan.

The reliability or internal consistency of the scale will be assessed by computing Cronbach's alpha coefficient of the items on the Likert Scale. Cronbach's alpha coefficient is a measure of correlation between the items in the scale, and values of 0.70 and above are considered adequate. Items which have low item-total correlation, or which strongly reduce the alpha total, will be revised or deleted in order to obtain a more homogeneous and internally consistent measure. The measure will not be used until a satisfactory Cronbach's alpha value is achieved, indicating that it consistently produces the same results.



Table 5. Cronbach's Alpha Interpretation

CRONBACH'S α	INTERNAL CONSISTENCY
0.90 and above	Excellent
0.80 - 0.89	Good
0.70 - 0.79	Acceptable
0.60 - 0.69	Questionable
0.50 - 0.59	Poor
below 0.50	Unacceptable

Source: Cronbach's George & Mallery (2003)

To validate the reliability of the system, the system-based approach will be conducted multiple times per project using the three selected Cagayan Provincial Capitol, Provincial Engineer's Office, Construction Division projects. A set of users will be asked to repeat the same operations over several weeks (e.g., log progress, generate S-curve, Bar chart, and receive email notifications) and check whether the system gets the same output and sends the email notifications under the same conditions. Any discrepancies in data processing or report generation will be logged, corrected, and retested until the workflow consistently produces the expected results.



3.7 Statistical Tools

The collected data from the Likert-scale questionnaire and the system logs will be analyzed using standard statistical tools appropriate for quantitative, survey-based research. The main statistical tools that will be used are:

Mean - Mean will be used to establish an overall rating for each question from respondents that will evaluate the usability, reliability, functionality, and performance efficiency of the system.

Formula: $\bar{x} = (w_1x_1 + w_2x_2 + \dots + w_nx_n) / (w_1 + w_2 + \dots + w_n)$ or

$$\bar{x} = \Sigma(w \times x) / \Sigma w$$

where: $\bar{x}$ = weighted mean

w = weight for each data point

x = the value itself

$w \times x$ = weight multiplied by value

$\Sigma(w \times x)$ = add up all the (weight $\times$ value) products

Σw = add up all the weights

Standard deviation - Standard deviation will be used to illustrate how much variance there is from the mean rating for each question.



Formula: $s = \sqrt{[\Sigma(x - \bar{x})^2 / (n - 1)]}$

where: s = sample standard deviation

x = each data value

$\bar{x}$ = sample mean (average)

n = number of data points

Inferential statistics (Wilcoxon Signed-Rank) - The Wilcoxon Signed-Rank Test will be utilized to determine whether there is a significant difference between two related sets of observations before and after the implementation of the automated system. The study is suitable to be tested using this nonparametric test because the data used in this study are ordinal in nature. The result of the test will show whether the changes in system performance observed are statistically significant.

Formula: $W = \min(W+, W-)$

where: W = the Wilcoxon test statistic, which is the smaller of the sums of the positive and negative ranks.

$W+$ = the sum of the positive ranks.

$W-$ = the sum of the negative ranks.

Cohen's d (Effective size) - Cohen's d will be used to measure the effect size or magnitude of the difference between two conditions, such as before and after system



implementation. It will indicate the magnitude of the difference between the two conditions and help determine the practical impact of the developed system. This will complement the Wilcoxon Signed-Rank Test by providing information on the magnitude of the observed changes in usability, reliability, functionality, and performance efficiency.

Table 6. Cohen's d Interpretation

0.20	Small effect	practical impact is minimal and may not be readily observable in applied settings
0.50	Medium size effect	moderate and meaningful difference that is likely to have practical relevance
0.80	Large size effect	substantial difference with strong practical significance

Correlation Analysis (Spearman's Rho) - Spearman's Rho will be used to establish the strength and direction of the relationship between system evaluation variables such as usability, reliability, functionality, and performance efficiency. This statistical tool is suitable for the study since the data to be collected are ordinal in nature and will be obtained using Likert-scale responses. The results of the correlation analysis will provide an understanding of the level of relationship between the variables, thus contributing to the assessment of the developed system.

The statistical analysis for this study will be carried out using the Statistical Package for the Social Sciences (SPSS), a widely used software for survey-based research in the social and applied sciences. The analysis will not require any unique or specialized



modeling equations because the Likert-scale questionnaires will be evaluated using established survey analysis methods.



3.8 Ethical Consideration

During the planning, execution, and evaluation stages of the study, ethical considerations will be properly adhered to by the researchers. The researchers will write a letter to the head of the Cagayan Provincial Capitol, Provincial Engineer's Office, Construction Division for this research study, as well as for the inclusion of personnel as respondents in the survey.

For each respondent who takes part in the study, a consent form must be completed prior to taking the survey. The researchers will explain the purpose of the study to ensure that respondents voluntarily participate in the study and that they may withdraw from the study at any time. Furthermore, the researchers will indicate to the participants the time required to complete the study and ensure that the participants understand that their participation or non-participation will not positively or negatively affect any evaluations or their employment status.

In order to ensure the confidentiality of each of the respondents, any personally identifiable information will be removed from any data reports. Specifically, any personally identifiable information, such as the names and positions of the respondents, will be coded and not included in the final thesis report. All data collected will be kept in a secure data facility and will only be accessed by the researchers and the adviser, with such handling following the guidelines set forth in the Data Privacy Act of 2012 (RA 10173), with particular attention to confidentiality, security, and lawful processing of personal information.



The researchers will take into account the potential risks such as bias, discrimination, and misuse of data when developing the system. To avoid data discrimination, for instance, only those with access to the project will be able to view the status of project updates. Furthermore, only appropriate stakeholders will receive automated e-mail notifications that contain only the project status updates. These e-mail notifications will not contain any confidential or sensitive personal or financial data, and all such data will be contained within the Cagayan Provincial Capitol, Provincial Engineer's Office, Construction Division's internal reporting system.

In order to avoid plagiarism in the research study, the researchers will ensure that information from existing studies is properly cited in the literature and methodology sections of the study. Furthermore, any existing survey, questionnaire, process, or tools that are to be utilized in the study will be described, and the authors of those tools will be credited accordingly.



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APPENDICES

Appendix A: Pre-test Questionnaire

Instruction: Please check (✓) the appropriate answer.

Role in the organization

- ☐ Contractor
- ☐ Engineer I
- ☐ Engineer II
- ☐ Engineer III
- ☐ Engineer IV

Sex

- ☐ Male
- ☐ Female

Age

- ☐ 18 – 24
- ☐ 25 – 34
- ☐ 35 – 44
- ☐ 45 and above

Working Experience

- ☐ 1 year or below
- ☐ 2 – 5 years
- ☐ 6 and above

Statement	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
1. Usability					
1.1 I expect that I will use ConstructFlow frequently.					
1.2 I expect this system may be more complex than necessary.					
1.3 I expect ConstructFlow to be easy to use.					
1.4 I expect that I would need assistance from a technical person to use ConstructFlow.					
1.5 I expect the different functions of ConstructFlow to work well together.					



1.6 I expect that there may be inconsistencies in ConstructFlow.					
1.7 I expect most people would learn to use ConstructFlow quickly.					
1.8 I expect ConstructFlow to be complicated to use.					
1.9 I expect that I would feel confident using ConstructFlow.					
1.10 I expect that I would need to learn many things before I could use ConstructFlow effectively.					
2. Reliability					
2.1 I expect that the ConstructFlow will be available at any time.					
2.2 I expect that I will rarely experience crashes, lag, or system failures while using ConstructFlow.					
2.3 I expect that the ConstructFlow will be easy to use on any operating system such as Windows and macOS.					
2.4 I expect that ConstructFlow will recover and continue to function properly if an error occurs.					
2.5 I expect that most faults in ConstructFlow can be eliminated over time.					



2.6 I expect that ConstructFlow can handle errors.					
2.7 I expect that ConstructFlow can resume operation and restore data.					
2.8 I expect that ConstructFlow can meet existing reliability standards.					
2.9 Overall, I expect ConstructFlow to be reliable.					
3. Functionality					
3.1 I expect that the information and data available in the ConstructFlow will be adequate.					
3.2 I expect ConstructFlow menu buttons to be functional and easy to use.					
3.3 I expect that the functions of the Constructflow web-based system will work well overall.					
3.4 I expect that the information and data available in the ConstructFlow will be comprehensive.					
3.5 I expect that ConstructFlow can perform the tasks assigned.					
3.6 I expect that ConstructFlow can produce the expected results.					
3.7 I expect that ConstructFlow can interact with other software or systems.					



3.8 I expect that ConstructFlow is equipped with acceptable security measures.					
3.9 I expect that ConstructFlow can meet existing requirements.					
3.10 Overall, I expect ConstructFlow to be very useful.					
4. Performance Efficiency					
4.1 I expect that ConstructFlow will respond quickly when displaying information.					
4.2 I expect that ConstructFlow will respond quickly when updating user information.					
4.3 I expect that ConstructFlow will process submitted data within an acceptable amount of time.					
4.4 I expect that ConstructFlow will use system resources efficiently when performing its functions.					
4.5 I expect that ConstructFlow will handle the required amount of data and user activities without affecting its performance.					

Source: Merged and adapted from Begier and Wdowicki (2007), Lusiani and Princes (2024), Cailing (2024), and Leyran-Amper and Delgado (2016).



APPENDICES

Appendix B: Post-test Questionnaire

Instruction: Please check (✓) the appropriate answer.

Role in the organization

- ☐ Contractor
- ☐ Engineer I
- ☐ Engineer II
- ☐ Engineer III
- ☐ Engineer IV

Sex

- ☐ Male
- ☐ Female

Age

- ☐ 18 – 24
- ☐ 25 – 34
- ☐ 35 – 44
- ☐ 45 and above

Working Experience

- ☐ 1 year or below
- ☐ 2 – 5 years
- ☐ 6 and above

Statement	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
1. Usability					
1.1 I would like to use ConstructFlow frequently.					
1.2 I found ConstructFlow to be more complex than necessary.					
1.3 I found ConstructFlow easy to use.					
1.4 I would need support from a technical person to use ConstructFlow.					



1.5 I found the various functions in ConstructFlow were well integrated.					
1.6 I found there was too much inconsistency in ConstructFlow.					
1.7 Most people would learn to use ConstructFlow very quickly.					
1.8 I found ConstructFlow to be very complicated to use.					
1.9 I feel very confident using ConstructFlow.					
1.10 I needed to learn a lot of things before I could get started with ConstructFlow.					
2. Reliability					
2.1 The ConstructFlow is available at any time.					
2.2 I rarely experience crashes, lag, or system failures while using ConstructFlow.					
2.3 The ConstructFlow can be easily used on any operating system such as Windows and macOS.					
2.4 ConstructFlow recovers and continues to function properly if an error occurs.					
2.5 Most faults in ConstructFlow can be eliminated over time.					



2.6 ConstructFlow can handle errors.					
2.7 ConstructFlow can resume operation and restore data.					
2.8 ConstructFlow can meet existing reliability standards.					
2.9 Overall, I consider ConstructFlow to be reliable.					
3. Functionality					
3.1 The information and data available in the ConstructFlow are adequate.					
3.2 The ConstructFlow menu buttons are functional and easy to use.					
3.3 The overall functions of the ConstructFlow web-based system work well.					
3.4 The information and data available in the ConstructFlow are comprehensive.					
3.5 ConstructFlow can perform the tasks assigned.					
3.6 ConstructFlow can produce the expected results.					
3.7 ConstructFlow can interact with other software or systems.					
3.8 ConstructFlow is equipped with acceptable security measures.					
3.9 ConstructFlow can meet existing requirements.					



3.10 Overall, ConstructFlow is very useful.					
4. Performance Efficiency					
4.1 ConstructFlow can respond quickly when displaying information.					
4.2 ConstructFlow can respond quickly when updating user information.					
4.3 ConstructFlow can process submitted data within an acceptable amount of time.					
4.4 ConstructFlow can use system resources efficiently when performing its functions.					
4.5 ConstructFlow can handle the required amount of data and user activities without affecting its performance.					

Source: Merged and adapted from Begier and Wdowicki (2007), Lusiani and Princes (2024), Cailing (2024), and Leyran-Amper and Delgado (2016).